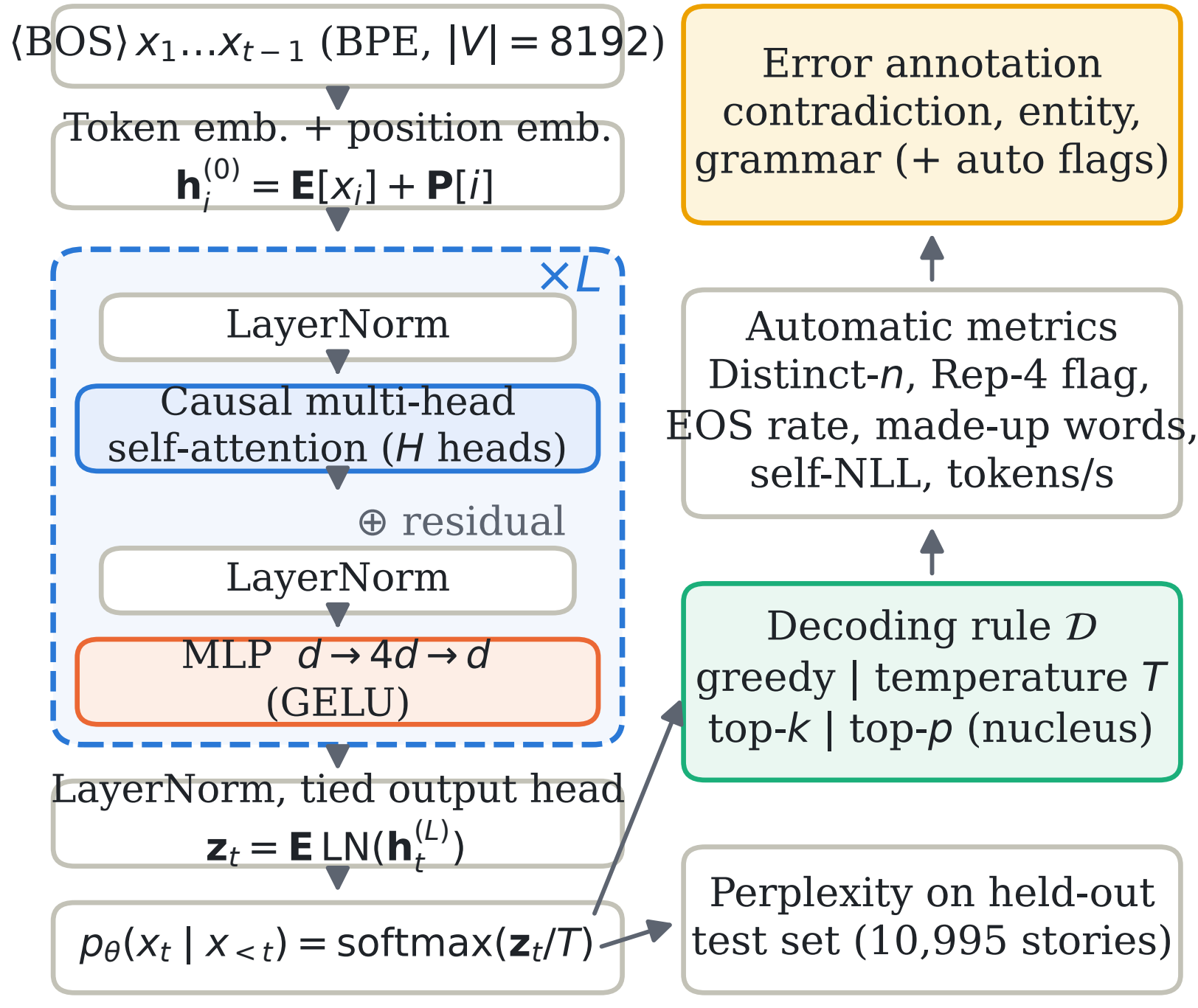
⟨BOS⟩ $x_1 \ldots x_{t-1}$ (BPE, $|V| = 8192$)
Token emb. + position emb.
$\mathbf{h}_i^{(0)} = \mathbf{E}[x_i] + \mathbf{P}[i]$
×L
LayerNorm
Causal multi-head self-attention (H heads)
⊕ residual
LayerNorm
MLP $d \to 4d \to d$ (GELU)
LayerNorm, tied output head
$\mathbf{z}_t = \mathbf{E}\,\mathrm{LN}(\mathbf{h}_t^{(L)})$
$p_\theta(x_t \mid x_{<t}) = \mathrm{softmax}(\mathbf{z}_t / T)$
Error annotation contradiction, entity, grammar (+ auto flags)
Automatic metrics Distinct-n, Rep-4 flag, EOS rate, made-up words, self-NLL, tokens/s
Decoding rule $\mathcal{D}$ greedy | temperature T top-k | top-p (nucleus)
Perplexity on held-out test set (10,995 stories)